Task # 1

```
#include <iostream>
using namespace std;
int main () {
double num;
cout << " enter a num ";
cin >> num;
if (num > 0)
{
cout << " num << "
        positive " << endl;
}
else
{
cout << "Number is negative" << endl
}
return 0;
}
```

number enter

Task #2

```
#include <iostream>
using namespace std;
int main(){
int num;
cout << "enter an integer";
cin >> num;
if (num > 0)
cout << "positive number" << endl;
else if (num < 0)
cout << "negative number" ;<< endl;
else
  cout << "zero" << endl;
return 0;
}
```

Task # 3

```
#include <iostream>
using namespace std;
int main() {
int marks;
cout << "enter marks";
cin >> marks;
if(marks >= 75)
{
cout << "Grade A" << endl;
}
else
{
  if (marks >= 60)
  {
  cout << "Grade B" << endl;
  }
else
{
  if(marks >= 45)
   {
  cout << "Grade C" << endl;
  }
 else
 {
  if(marks < 30)
  {
  cout << "Grade D" << endl;
  }
  else
  {
  cout << "
```

Task # 4(a)

```
#include <iostream>
using namespace std;
int main ( ){
int a, b, c, largest;
cout << "enter three numbers";
cin >> a >> b >> c;
if (a >= b)
{
  if (a >= c)
   largest = a;
   else
     largest = c;
  }
  else
  {
    if (b >= c)
      largest = b;
     else
     largest = c;
  }
  cout << "largest number = "largest <<
  return 0;
  }
```

Task #4 (b)

```
#include <iostream>
using namespace std;
int main (){
int n, digits;
cout << "enter a number ";
cin >> n;
if (n>0)
{ if (n<10)
else if (n<100)
else if (n<1000)
else if (n<10000)
else if (n<100000)
else if (n<1000000)
else if (n<10000000)
else if (n<100000000)
else if (n<1000000000)
else
cout << n << " is positive and has " << c
    << "digit(s)" << endl;
  }
  else
  {
  cout << "number is not positive, so c
  are not counted" << endl;
  }
  return 0;
}
```

Lab 4 - Task #1

Code

(invalid)* **new*** new* new new

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5   double num;
6   cout << "Enter a number: ";
7   cin >> num;
8   if (num > 0)
9   {
10  cout << num << " - number entered is
    positive" << endl;
11  }
12  else
13  {
14  cout << "Number is negative" << endl;
15  }
16  return 0;
17 }
```

Output

```
Enter a number: 15
15 - number entered is positive
```

Sample run of the same code

Lab 4 - Task #2

Code

(invalid)* new* new* new new

```cpp
#include <iostream>
using namespace std;
int main()
{
 int num;
 cout << "Enter an integer: ";
 cin >> num;
 if (num > 0)
 cout << "Positive number" << endl;
 else if (num < 0)
 cout << "Negative number" << endl;
 else
 cout << "Zero" << endl;
 return 0;
}
```

Output

```
Enter an integer: -7
Negative number
```

Sample run of the same code

Lab 4 - Task #3

Code

(invalid)* new* new* new new

```cpp
#include <iostream>
using namespace std;
int main()
{
 int mark;
 cout << "Enter mark: ";
 cin >> mark;
 if (mark >= 75)
 {
 cout << "Grade: A" << endl;
 }
 else
 {
 if (mark >= 60)
 {
 cout << "Grade: B" << endl;
 }
 else
 {
 if (mark >= 45)
 {
 cout << "Grade: C" << endl;
 }
 else
 {
 if (mark < 30)
 {
 cout << "Grade: D" << endl;
 }
 else
 {
 cout << "No grade defined for 30-44" <<
endl;
 }
```

Output

```
Enter mark: 68
Grade: B
```

Sample run of the same code

Lab 4 - Task #4(a)

Code

(invalid)* new* new* new new

```cpp
#include <iostream>
using namespace std;
int main()
{
 int a, b, c, largest;
 cout << "Enter three numbers: ";
 cin >> a >> b >> c;
 if (a >= b)
 {
 if (a >= c)
 largest = a;
 else
 largest = c;
 }
 else
 {
 if (b >= c)
 largest = b;
 else
 largest = c;
 }
 cout << "Largest number = " << largest << endl;
 return 0;
}
```

Output

```
Enter three numbers: 12 45 30
Largest number = 45
```

Sample run of the same code

Lab 4 - Task #4(b)

Code

(invalid)* new* new* new new

```
2  using namespace std;
3  int main()
4  {
5   int n, digits;
6   cout << "Enter a number: ";
7   cin >> n;
8   if (n > 0)
9   {
10  if (n < 10) digits = 1;
11  else if (n < 100) digits = 2;
12  else if (n < 1000) digits = 3;
13  else if (n < 10000) digits = 4;
14  else if (n < 100000) digits = 5;
15  else if (n < 1000000) digits = 6;
16  else if (n < 10000000) digits = 7;
17  else if (n < 100000000) digits = 8;
18  else if (n < 1000000000) digits = 9;
19  else digits = 10;
20  cout << n << " is positive and has " <<
    digits << " digit(s)" << endl;
21  }
22  else
23  {
24  cout << "Number is not positive, so digits
    are not counted" << endl;
25  }
26  return 0;
27 }
```

Output

```
Enter a number: 4821
4821 is positive and has
4 digit(s)
```

Sample run of the same code